Eshaal Assigned Source Review

Operating Systems CEP Topic 23 | 22 October 2026

Eight report copies, seven unique studies. First source review prepared at Raazia’s request; human independent signoff remains pending.

Item	Reviewed	Remaining
Primary-study QA items	70	Independent adjudication
Report eligibility criteria	48	Human independent checks
Retained outcomes	30	Independent source checks
Methodology audit findings	14	Coverage/provenance issues listed below

Original QA scores and frozen baseline are preserved. The source-based review verifies retained values and identifies interpretation/version corrections. It is not a rerun of experiments or a complete census of all screening decisions.

The three packages now cover all20included studies. Master integration is complete. The20study target is met; independent human checks,559record selection and historical search limitations remain open.

Main corrections

ES01 / P001 E001 E002: PDFpp16-17 / printed458-459; GPU-specific TensorRT/Poisson context retained. Source: Figure14 and section6.7

ES02 / P002: Final typeset TPDS copy inspected; correct Xplore article identifier10002315; retain source URL/hash. Source: Final DOI/title/page1 and IEEE metadata

ES03 / P004: Retain under separable single-GPU inference evidence; Table3 is added mean overhead,not P99/total latency. Source: PDFpp11-12

ES04 / P005 P023: SameS005 per explicit extension statement; conferenceAMD results distinct from journalNVIDIA/LLM additions. Source: P023p1;pp23-24,31-33

ES05 / P023: Keep implementations separate; TVM0.8/Ubuntu18.04/16GBhost versus TVM0.14/Ubuntu20.04/32GBhost. Source: P005p11;P023p23

ES06 / P003 E006: CDF across co-location scenarios; no requestP90/P99 inference. Source: PDFp6;Figure7

ES07 / P006 E028: Added empty-kernel instrumentationP90 overhead at160blocks with aggregation. Source: PDFp12;Figure15

ES08 / P007: Same trust domain; reconstructedREEF-N comparator and aggregate throughput conditions retained. Source: PDFpp9,12-13

ES09 / All studies Q9: Retain assessed scores in Eshaal package; flag team calibration before final master totals. Source: SubmittedQ9;Raazia/Rameen proposals;source anchors

ES10 / P002 P003 P005: Q9=2 for direct claim-specific performance interference tests; P003 Q10=2 for versioned artifact appendix. Original scores remain visible.. Source: Source anchors in QA rows; common Q9 rubric applied without extrapolating to security/fault isolation

S001 / P001 - VastPipe

Version: Journal typeset copy with cover; article startsPDFp2

Source: https://jcst.ict.ac.cn/en/article/pdf/preview/10.1007/s11390-024-3773-5.pdf

DOI/proceedings: 10.1007/s11390-024-3773-5

PDF pages: 21. SHA256: 38451986cdaeb1f903822f9b30969b2b08ca652bfd487838ebefe020e7ef5552

Criterion	Finding and source anchor
IC1	English peer-reviewed report identity matched to title,DOI/proceedings and year. Journal typeset copy with cover; article startsPDFp2
IC2	Local sharing mechanism inspectable. PDF pp16-17; sections6.7-6.8
IC3	Inference workload identified; mixed training cases retained only for separable inference. PDF pp16-17; Figures14-16
IC4	Relevant measured outcomes and methods inspectable. PDF p17 / printed459; Figure14 onp16
IC5	Within-one-GPU evidence distinguishable from cluster outcomes. PDF pp16-17; Figures14-16
IC6	Publication year lies in2020-2026 main window. Title/metadata

Quality appraisal

Original QA: 13/18; current appraised QA: 13/18. Source-based calibration is logged; independent adjudication remains pending.

Item	Score	Rationale and anchor
Q1	2	Local GPU portability and throughput objective are explicit; NPU cluster results are excluded from this extracted stratum. PDF pp2,16-17; section6.7
Q2	2	RL-tuned MPS allocation and MI100 CU-mask backend are described. PDF pp16-17; sections6.7-6.8
Q3	1	GPU throughput provenance is clear, but GPU-specific repeated-run uncertainty is not established. PDF pp16-18
Q4	1	A100/TensorRT and MI100/TVM/ROCm are named; exact versions, GPU batch/rate and tuned quotas remain incomplete. PDF pp16-17
Q5	1	Same A100 no/default/tuned MPS comparison; default means unspecified respective quotas. PDF pp16-17; Figure14
Q6	2	2.7x versus noMPS and1.5x versus defaultMPS are explicit for VGG16+GoogLeNet; not universal averages. PDF p17 / printed459; Figure14 onp16
Q7	N/A	No online latency claim in the retained local GPU outcomes; NPU results cannot fill this item. PDF pp16-17; sections6.7-6.8
Q8	2	Several paired CV/NLP model co-locations are measured. PDF pp16-17; Figures14-16
Q9	1	Performance sharing is measured; separate security/fault validation is not established. Q9 calibration remains flagged. PDF pp16-17
Q10	1	Stable-mix Q-table assumption and retuning for changed demand are explicit; GPU replication detail remains incomplete. PDF p17; section6.9

Checked retained outcomes

Evidence	Value and metric	Context and anchor
E001	2.7 x; Throughput ratio	NVIDIA A100. VGG16+GoogLeNet. Baseline: No MPS. PDF pp16-17 / printed458-459; Figure14; section6.7
E002	1.5 x; Throughput ratio	NVIDIA A100. VGG16+GoogLeNet. Baseline: Default MPS. PDF pp16-17 / printed458-459; Figure14; section6.7

2.7x and1.5x match paired-model prose. TensorRT and Poisson model-count description are available; exact rate,batch and tuned quotas remain unresolved. Abstract/global headline is not substituted.

S002 / P002 - iGniter

Version: Final IEEE TPDS34(3),812-827,2023 typeset copy; matched DOI and retained values

Source: https://prefer-nsf.org/pdf/IEEE%20TPDS%20_%20igniter%202023.pdf

DOI/proceedings: 10.1109/TPDS.2022.3232715

PDF pages: 16. SHA256: 52a6e65fcdc3129dea7d878a089dc7d1c151b6f7ee08e4cd845e22598ca66f79

Criterion	Finding and source anchor
IC1	English peer-reviewed report identity matched to title,DOI/proceedings and year. Final IEEE TPDS34(3),812-827,2023 typeset copy; matched DOI and retained values
IC2	Local sharing mechanism inspectable. Final PDF pp8-10; section4.2
IC3	Inference workload identified; mixed training cases retained only for separable inference. Final PDF p11; Figure13
IC4	Relevant measured outcomes and methods inspectable. Final PDF pp10-13
IC5	Within-one-GPU evidence distinguishable from cluster outcomes. Final PDF p11; Figure13
IC6	Publication year lies in2020-2026 main window. Title/metadata

Quality appraisal

Original QA: 17/20; current appraised QA: 18/20. Source-based calibration is logged; independent adjudication remains pending.

Item	Score	Rationale and anchor
Q1	2	Predictable cost-efficient inference provisioning is the stated objective. Final PDF p1
Q2	2	MPS active-thread allocation, placement and shadow-Triton recovery are specified. Final PDF pp8-10; section4.2
Q3	2	Model validation repeats three times with SD bars; resource plan and P99 observation windows are described. Final PDF pp10-12; sections5.2-5.3
Q4	1	V100 instances, four models and batch/arrival settings are given; exact OS/driver coverage remains incomplete. Final PDF p10; Table3; section5.1
Q5	1	GSLICE+ and gpulets+ are modified; batch/allocation objectives differ and modifications must remain visible. Final PDF p10; Baselines and Metrics
Q6	2	Cost, allocation and throughput are separately defined; cluster cost is not local GPU throughput. Final PDF pp10-13
Q7	2	P99 over e.g.30seconds and first-second15.6ms versus10ms SLO, then shadow activation at1.5seconds are explicit. Final PDF pp11-12; section5.3; Figure17
Q8	2	Four co-located models with25% allocation and batch3 test local interference. Final PDF p11; Figure13
Q9	2	Q9=2 for direct measured performance-interference evaluation within the stated service scope; this does not establish security or fault isolation. PDF pp8-10; interference/provisioning evaluation
Q10	2	Profiling cost and PCIe/prediction assumptions are disclosed; source code identified in final article. Final PDF pp11,13-14; sections5.4,6

Checked retained outcomes

Evidence	Value and metric	Context and anchor
E003	1.53-5.02 percent; Prediction error range	NVIDIA V100. AlexNet/ResNet50/VGG19/SSD; batch3. Baseline: Measured local latency. Final PDF p11 / printed822; Figure13
E004	15.6 ms; Initial p99 latency	NVIDIA V100. W1 with W5 and W11. Baseline: W1 latency SLO10ms. Final PDF p12 / printed823; Figure17; section5.3

Prediction error range and transient P99 violation verified in final typeset copy; distinguish arXiv page offsets and final printed pages.

S003 / P003 - gpulets

Version: USENIX ATC2022 proceedings with cover

Source: https://www.usenix.org/system/files/atc22-choi-seungbeom.pdf

DOI/proceedings: USENIX proceedings (no DOI assigned)

PDF pages: 18. SHA256: 8b3ec8c3ac38f3d1fbe29361e65a6f8afee5ae8f9fca57b1e30ff15466027a9f

Criterion	Finding and source anchor
IC1	English peer-reviewed report identity matched to title,DOI/proceedings and year. USENIX ATC2022 proceedings with cover
IC2	Local sharing mechanism inspectable. PDF pp7-10; sections4-5
IC3	Inference workload identified; mixed training cases retained only for separable inference. PDF p6; section3.4
IC4	Relevant measured outcomes and methods inspectable. PDF p11; section5.2
IC5	Within-one-GPU evidence distinguishable from cluster outcomes. PDF p6; section3.4
IC6	Publication year lies in2020-2026 main window. Title/metadata

Quality appraisal

Original QA: 18/20; current appraised QA: 20/20. Source-based calibration is logged; independent adjudication remains pending.

Item	Score	Rationale and anchor
Q1	2	SLO-aware spatio-temporal sharing objective is explicit. PDF pp2-3
Q2	2	MPS gpulet partitioning, scheduling, interference model and process replacement are described. PDF pp7-10; sections4-5
Q3	2	Poisson arrivals and three repetitions with median result are explicit. PDF pp10-11; section5.1
Q4	2	CPU,RTX2080Ti,OS,driver,CUDA,PyTorch and models are tabulated. PDF p10; Tables2-3
Q5	2	Ported SBP and greedy best-fit conditions, and why full Nexus is not compared, are explained. PDF p10; section5.1
Q6	2	SLO-preserved throughput is maximum rate with99% requests within SLO; multi-GPU results not substituted for local CDF. PDF p11; section5.2
Q7	2	17.66% is the90%-of-scenarios overhead threshold, not requestP90/P99. PDF p6 / printed203; Figure7
Q8	2	250 model/batch combinations across partition ratios test local interference. PDF p6; section3.4
Q9	2	Q9=2 for direct measured interference and allocation-boundary evaluation; no hostile-tenant security guarantee is inferred. PDF pp3-4,8-12; sections3 and5
Q10	2	Artifact appendix provides versioned dependencies, Docker/build/run instructions and limitations; availability/replicability reporting is scored without claiming a local replication. Zenodo 10.5281/zenodo.6544909 and casys-kaist/glet are supplied. PDF pp17-18; artifact appendix: Ubuntu18.04/kernel4.15, LibTorch1.10, CUDA>=10.2, cuDNN>=7.6, Boost>=1.6, OpenCV>=4, CMake>=3.19

Checked retained outcomes

Evidence	Value and metric	Context and anchor
E006	17.66 percent overhead below threshold; Distribution threshold:90% of measured pairs	RTX2080Ti11GB. Paired model inferences. Baseline: Solo-running latency. PDF p6 / printed203; Figure7; section3.4

90% refers to model/batch/partition scenarios, not request-tail percentile.17.66% plotted threshold matches retained value.

S004 / P004 - PipeSwitch

Version: USENIX OSDI2020 proceedings with cover

Source: https://www.usenix.org/system/files/osdi20-bai.pdf

DOI/proceedings: USENIX proceedings (no DOI assigned)

PDF pages: 17. SHA256: bda339767febf69c5aaf9091fbfef42f3073df27c7f1a30b8973dd7867e7c0aa

Criterion	Finding and source anchor
IC1	English peer-reviewed report identity matched to title,DOI/proceedings and year. USENIX OSDI2020 proceedings with cover
IC2	Local sharing mechanism inspectable. PDF pp6-11; section4-5
IC3	Inference workload identified; mixed training cases retained only for separable inference. PDF p11; Workloads and section6.1
IC4	Relevant measured outcomes and methods inspectable. PDF pp12-13; Figure6
IC5	Within-one-GPU evidence distinguishable from cluster outcomes. PDF p11; Workloads and section6.1
IC6	Publication year lies in2020-2026 main window. Title/metadata

Quality appraisal

Original QA: 17/20; current appraised QA: 17/20. Source-based calibration is logged; independent adjudication remains pending.

Item	Score	Rationale and anchor
Q1	2	Time-sharing and fast context-switching between training/inference tasks are explicit. PDF p2; abstract
Q2	2	Pipelined transmission, controller, daemon and active/standby workers are specified. PDF pp6-11; section4-5
Q3	2	100-run means and Figure6 min/max are explicitly reported. PDF p11; Metrics
Q4	1	V100/T4,PCIe,memory,PyTorch1.3.0 andCUDA10.1 given; exact OS/kernel/driver unresolved. PDF p11; Setup
Q5	2	Ready-model baseline, same modified PyTorch and MPS full-memory/unified-memory swapping condition are explicit. PDF pp11-12; section6.1
Q6	2	Utilization is throughput relative to ready-model upper bound; eight-instance scheduling-cycle result kept distinct from local table. PDF pp12-13; Figure6
Q7	2	Table3 subtracts ready-model latency; it reports added mean overhead, not total latency or P99. PDF pp11-12; Table3
Q8	2	Single-GPU training-preempted-by-inference workload has separate inference latency; qualifies under mixed-workload rule. PDF p11; Workloads and section6.1
Q9	1	Process/memory control is described, but independent hostile-client/fault experiment is not established. PDF pp8-11
Q10	1	Stack versions, model registration and checkpoint dependence supplied; full reproduction configuration remains incomplete. PDF pp10-11; sections5-6

Checked retained outcomes

Evidence	Value and metric	Context and anchor
E005	6.01 ms; Additional end-to-end latency	NVIDIA V10016GB. ResNet152 inference batch8 + training batch32. Baseline: Ready-model latency subtracted; MPS overhead307.02ms. PDF pp11-12 / printed508-509; Table3; section6.1
E007	307.02 ms; Additional end-to-end latency	NVIDIA V10016GB. ResNet152 inference batch8 + training batch32. Baseline: Ready-model latency subtracted. PDF pp11-12 / printed508-509; Table3; section6.1
E008	5.40 ms; Additional end-to-end latency	NVIDIA V10016GB. Inception_v3 inference batch8 + training batch32. Baseline: Ready-model latency subtracted. PDF pp11-12 / printed508-509; Table3; section6.1

Evidence	Value and metric	Context and anchor
E009	232.25 ms; Additional end-to-end latency	NVIDIA V10016GB. Inception_v3 inference batch8 + training batch32. Baseline: Ready-model latency subtracted. PDF pp11-12 / printed508-509; Table3; section6.1
E010	10.27 ms; Additional end-to-end latency	NVIDIA V10016GB. Bert_base inference batch8 + training batch32. Baseline: Ready-model latency subtracted. PDF pp11-12 / printed508-509; Table3; section6.1
E011	204.52 ms; Additional end-to-end latency	NVIDIA V10016GB. Bert_base inference batch8 + training batch32. Baseline: Ready-model latency subtracted. PDF pp11-12 / printed508-509; Table3; section6.1
E012	5.57 ms; Additional end-to-end latency	NVIDIA T416GB. ResNet152 inference batch8 + training batch32. Baseline: Ready-model latency subtracted. PDF pp11-12 / printed508-509; Table3; section6.1
E013	259.20 ms; Additional end-to-end latency	NVIDIA T416GB. ResNet152 inference batch8 + training batch32. Baseline: Ready-model latency subtracted. PDF pp11-12 / printed508-509; Table3; section6.1
E014	7.66 ms; Additional end-to-end latency	NVIDIA T416GB. Inception_v3 inference batch8 + training batch32. Baseline: Ready-model latency subtracted. PDF pp11-12 / printed508-509; Table3; section6.1
E015	193.05 ms; Additional end-to-end latency	NVIDIA T416GB. Inception_v3 inference batch8 + training batch32. Baseline: Ready-model latency subtracted. PDF pp11-12 / printed508-509; Table3; section6.1
E016	34.56 ms; Additional end-to-end latency	NVIDIA T416GB. Bert_base inference batch8 + training batch32. Baseline: Ready-model latency subtracted. PDF pp11-12 / printed508-509; Table3; section6.1
E017	338.25 ms; Additional end-to-end latency	NVIDIA T416GB. Bert_base inference batch8 + training batch32. Baseline: Ready-model latency subtracted. PDF pp11-12 / printed508-509; Table3; section6.1

All12 retained Table3 cells checked by model,GPU and mechanism.100-run added overhead after subtracting ready-model latency; MPS retains memory-full training and unified-memory swapping.

S005 / P005 - REEF conference

Version: USENIX OSDI2022 proceedings with cover; AMD results and restricted REEF-N

Source: https://www.usenix.org/system/files/osdi22-han.pdf

DOI/proceedings: USENIX proceedings (no DOI assigned)

PDF pages: 21. SHA256: d0c1dfc5274d4a44826d02b70d78a90b7b9f1279d87804f398a1386abcecce91

Criterion	Finding and source anchor
IC1	English peer-reviewed report identity matched to title,DOI/proceedings and year. USENIX OSDI2022 proceedings with cover; AMD results and restricted REEF-N
IC2	Local sharing mechanism inspectable. PDF pp4-11; sections3-6
IC3	Inference workload identified; mixed training cases retained only for separable inference. PDF pp11-12; Table2
IC4	Relevant measured outcomes and methods inspectable. PDF p12 / printed549; Figure11
IC5	Within-one-GPU evidence distinguishable from cluster outcomes. PDF pp11-12; Table2
IC6	Publication year lies in2020-2026 main window. Title/metadata

Quality appraisal

Original QA: 17/20; current appraised QA: 18/20. Source-based calibration is logged; independent adjudication remains pending.

Item	Score	Rationale and anchor
Q1	2	Concurrent RT/BE DNN inference with low latency and work conservation is explicit. PDF p2
Q2	2	Reset preemption, idempotent re-execution, queues and dynamic kernel padding are described. PDF pp4-11; sections3-6
Q3	1	DISB scenarios and baseline methods clear; end-to-end repeated-run/variance protocol not established. PDF pp11-12; sections7.1-7.2
Q4	2	MI50,CPU,16GB host RAM,Ubuntu18.04,ROCm4.3.0,TVM0.8.0 and clients supplied. PDF p11; Table2
Q5	2	RT-only,SEQ,GPUStreams and extended wait-based preemption conditions are explicit. PDF pp11-12; sections7.1-7.3
Q6	2	Figure11 throughput includes both RT and BE clients; benchmark/load is retained for each ratio. PDF p12 / printed549; Figure11
Q7	1	End-to-end RT overhead is stated with offered loads; request percentile for retained headline overheads unresolved. PDF pp11-12; sections7.1-7.2
Q8	2	Single/multiple RT and BE clients under uniform,Poisson and trace workloads are tested. PDF pp11-12; Table2
Q9	2	Q9=2 for explicitly defined priority/performance protection and direct latency-interference tests; journal-only NVIDIA results remain separate. Conference PDF pp2-3,8-11; sections2 and5
Q10	2	Artifact,platform and idempotence constraints disclosed; conference REEF-N is distinct from later journal NVIDIA implementation. PDF pp10-11,15-16; P023p1

Checked retained outcomes

Evidence	Value and metric	Context and anchor
E018	1.60 x; Overall throughput ratio	AMD MI5016GB. DISB A:1 VGG19 RT client100req/s uniform +1 ResNet152 BE. Baseline: RT-Only dedicated GPU. PDF pp11-12 / printed548-549; Table2; Figure11; section7.2
E019	0.5 percent; RT end-to-end latency overhead	AMD MI5016GB. DISB A:1 VGG19 RT client100req/s uniform +1 ResNet152 BE. Baseline: RT-Only dedicated GPU. PDF pp11-12 / printed548-549; Table2; Figure11; section7.2
E020	1.14 x; Overall throughput ratio	AMD MI5016GB. DISB B:1 VGG19 RT client220req/s uniform +1 ResNet152 BE. Baseline: RT-Only dedicated GPU. PDF pp11-12 / printed548-549; Table2; Figure11; section7.2

Evidence	Value and metric	Context and anchor
E021	1 percent; RT end-to-end latency overhead	AMD MI5016GB. DISB B:1 VGG19 RT client220req/s uniform +1 ResNet152 BE. Baseline: RT-Only dedicated GPU. PDF pp11-12 / printed548-549; Table2; Figure11; section7.2
E022	<1.5 percent; RT end-to-end latency overhead	AMD MI5016GB. DISB C-E:multiple BE clients. Baseline: RT-Only dedicated GPU. PDF pp11-12 / printed548-549; Table2; Figure11; section7.2
E023	3.00 x; Overall throughput ratio	AMD MI5016GB. DISB D:5 RT model clients20req/s uniform +5 BE model clients. Baseline: RT-Only dedicated GPU. PDF pp11-12 / printed548-549; Table2; Figure11; section7.2
E024	2.96 x; Overall throughput ratio	AMD MI5016GB. DISB E:5 RT model clients20req/s Poisson +5 BE model clients. Baseline: RT-Only dedicated GPU. PDF pp11-12 / printed548-549; Table2; Figure11; section7.2
E025	7.7 x; Overall throughput ratio	AMD MI5016GB. Apollo REAL trace about43 RTreq/s +5 BE model clients. Baseline: RT-Only dedicated GPU. PDF pp11-12 / printed548-549; Table2; Figure11; section7.2
E026	<2 percent; RT end-to-end latency overhead	AMD MI5016GB. Apollo REAL trace about43 RTreq/s +5 BE model clients. Baseline: RT-Only dedicated GPU. PDF pp11-12 / printed548-549; Table2; Figure11; section7.2

All nine values match scenario prose. RT+BE throughput,RT latency and load-specific denominator retained; journal results are not substituted.

S006 / P006 - Paella: Low-latency Model Serving with Software-defined GPU Scheduling

Version: Author-hosted SOSP2023 copy with conference/DOI

Source: https://vincen.tl/files/ng23paella.pdf

DOI/proceedings: 10.1145/3600006.3613163

PDF pages: 16. SHA256: 99d90b79293be180ddb888abf700583ddc530c62fcc2cca603b9a533e62f602e

Criterion	Finding and source anchor
IC1	English peer-reviewed report identity matched to title,DOI/proceedings and year. Author-hosted SOSP2023 copy with conference/DOI
IC2	Local sharing mechanism inspectable. PDF pp5-9; sections4-6
IC3	Inference workload identified; mixed training cases retained only for separable inference. PDF pp10-12; Figures12-13
IC4	Relevant measured outcomes and methods inspectable. PDF pp10-11; Figures11-12
IC5	Within-one-GPU evidence distinguishable from cluster outcomes. PDF pp10-12; Figures12-13
IC6	Publication year lies in2020-2026 main window. Title/metadata

Quality appraisal

Original QA: 16/18; current appraised QA: 16/18. Source-based calibration is logged; independent adjudication remains pending.

Item	Score	Rationale and anchor
Q1	2	Fine-grained software scheduling for low-latency inference is explicit. PDF p1
Q2	2	Instrumented kernel placement/completion notifications,dispatcher and SRPT/deficit policy are described. PDF pp5-9; sections4-6
Q3	1	Stabilization before measurement stated; repetitions and uncertainty for headline curve not established. PDF p10; Methodology
Q4	2	T4,CPU,RAM,Linux4.15,CUDA11.7,TVM0.10 and model list supplied. PDF pp9-10; Table2
Q5	1	TritonTVM wrapper,omitted Clockwork curves and MPS process-count restriction require qualifications. PDF p10; Table3; footnote3
Q6	2	Average throughput versus P99 curves use increasing load until saturation. PDF pp10-11; Figures11-12
Q7	2	Up-to3x short-job P99 reduction is author-reported;6.6us is instrumentationP90 overhead, not request latency. PDF pp10-12; Figures12,15
Q8	2	Mixed short/long jobs and deficit threshold show latency/fairness trade-offs. PDF pp10-12; Figures12-13
Q9	N/A	No separately appraised resource/security/fault isolation guarantee; retain reasoned N/A. PDF pp9-12; scheduling and discussion
Q10	2	Compiler modifications,versioned setup,CPU dispatch and dynamic-batching limitations described. PDF pp9-13; sections7-8

Checked retained outcomes

Evidence	Value and metric	Context and anchor
E027	up to3 x; ResNet18 p99 reduction factor	Tesla T4. ResNet18+InceptionV3;lognormal sigma1.5/2. Baseline: CUDA-MS. PDF p10 / printed604; section7.1; Figure12 onPDFp11
E028	6.6 microseconds; p90 instrumentation overhead	Tesla T4. Empty kernel;160 blocks;instrumentation with aggregation. Baseline: Uninstrumented kernel. PDF p12 / printed606; section7.3; Figure15

Up-to3x is short-job P99 claim;6.6us is added P90 instrumentation overhead for160blocks with aggregation,not request latency.

S007 / P007 - Orion: Interference-aware, Fine-grained GPU Sharing for ML Applications

Version: Author-hosted EuroSys2024 copy with conference/DOI

Source: https://fotstrt.github.io/files/2024-orion.pdf

DOI/proceedings: 10.1145/3627703.3629578

PDF pages: 18. SHA256: 7421b546dc81577d787b2c8e17709da7b560d4d2992fc67e5c0898adc8eb8ae7

Criterion	Finding and source anchor
IC1	English peer-reviewed report identity matched to title,DOI/proceedings and year. Author-hosted EuroSys2024 copy with conference/DOI
IC2	Local sharing mechanism inspectable. PDF pp5-8; sections4-5
IC3	Inference workload identified; mixed training cases retained only for separable inference. PDF pp9,12; sections6.2-6.3
IC4	Relevant measured outcomes and methods inspectable. PDF p12; section6.2.3
IC5	Within-one-GPU evidence distinguishable from cluster outcomes. PDF pp9,12; sections6.2-6.3
IC6	Publication year lies in2020-2026 main window. Title/metadata

Quality appraisal

Original QA: 19/20; current appraised QA: 19/20. Source-based calibration is logged; independent adjudication remains pending.

Item	Score	Rationale and anchor
Q1	2	Interference-aware fine-grained local GPU sharing is explicit. PDF p1
Q2	2	CUDA interception,per-client queues,resource profiling and throttling are specified. PDF pp5-8; sections4-5
Q3	2	Three experiment repetitions and interpretation of variance bars are specified. PDF pp8-9,12; sections6.1,6.3
Q4	2	V10016GB/A10040GB,VM,PyTorch1.12,Python3.9,CUDA10.2/11.3 and models supplied. PDF pp4,8-9; Tables1,3
Q5	1	REEF-N and Tick-Tock are reimplemented; Streams threads and MPS processes differ. PDF p9; Baselines
Q6	2	7.3x is aggregate throughput maximum versus GPU dedicated only to high-priority inference. PDF p12; section6.2.3
Q7	2	Poisson15% and Apollo22% overhead summaries are distinct; Figure12 averages across co-locations. PDF pp9,12; Figures11-12
Q8	2	Two-client tests and five-clientA100 experiment test interference at identified scales. PDF pp9,12; sections6.2-6.3
Q9	2	Same-trust-domain assumption explicitly limits isolation; direct performance evidence does not imply security. PDF p13; Security
Q10	2	Interception/profiling cost,cache interference and cluster co-design limits are explicit. PDF p13; sections6.5,7

Checked retained outcomes

Evidence	Value and metric	Context and anchor
E029	up to7.3 x; Aggregate throughput ratio	V10016GB;A10040GB extension. Two inference clients;Poisson;V100. Baseline: Dedicated high-priority inference. PDF p12 / printed1086; section6.2.3; Figure12
E030	15 percent; High-priority p99 overhead	V10016GB;A10040GB extension. Two inference clients;Poisson;V100. Baseline: Dedicated high-priority inference. PDF p12 / printed1086; section6.2.3; Figure12

7.3x is maximum aggregate throughput;15% is authors' Poisson high-priority latency summary over collocations,not worst-case/untrusted guarantee.

S005 / P023 - Real-time, Work-conserving GPU Scheduling for Concurrent DNN Inference

Version: Author-hosted42page journal typeset copy; publication footerNovember2025; extension statement onp1

Source: https://media.jryang.com/files/reef-tocs25.pdf

DOI/proceedings: 10.1145/3768622

PDF pages: 42. SHA256: 462d0be310b4f92c1230e6e7fc0a8fb6185b885d40230375777e3364418024b8

Criterion	Finding and source anchor
IC1	Journal extension ofP005 explicitly identified; no new study. PDF p1
IC2	Reset preemption and padding implemented on AMD and NVIDIA. PDF pp9,15-23
IC3	TVM,PyTorch,TensorRT DNN/LLM inference workloads. PDF pp23-24,31-33
IC4	Normalized latency/throughput metrics and comparisons inspectable. PDF pp24-25,31-32
IC5	OneMI50 and oneGV100 configurations separately evaluated. PDF p23; section6.2
IC6	Source footer givesNovember2025,within period; distinguish issue versus online dates. PDF pp1-2

Companion report linked toS005. No extra study-level QA form or new study count is created. Its new NVIDIA/LLM experiments must be extracted as separate report/version strata if used quantitatively.

The first-page extension statement links P023 to P005. Source setup: TVM0.14/Ubuntu20.04/32GB host and full NVIDIA implementation, versus conference TVM0.8/Ubuntu18.04/16GB host and restrictedREEF-N. Do not transplant journal performance into the conferenceAMD outcome rows. Journal Figures28-30 add Triton,Orion and LLM comparisons under distinct conditions.

Methodology audit and actions

MA01 Protocol preservation (Resolution documented)

Frozen protocol preserved; all later amendments and execution deviations are separate.

Evidence: task1/submitted_protocol.pdf; evidence/protocol_amendments.md

Next: Keep submitted protocol unchanged; source hash verified in integrated validation.

MA02 Counting arithmetic (Green)

4189-114=4075;3493+559+21+2=4075;21reports=20studies

Evidence: baseline_PRISMA_Counts.csv; Screening_Log; Duplicate_Removal

Next: Arithmetic verified; this does not validate every screening judgment

MA03 Duplicate linkage (Green)

114unique removed IDs;all114canonical targets exist in4075record register

Evidence: baseline_Duplicate_Removal.csv;baseline_Screening_Log.csv

Next: Keep original links;pairwise semantic validation is a separate task

MA04 559-record stopping rule (Explicit limitation)

559 retained records were not advanced; their full eligibility is unknown. The 20-study count does not close this branch.

Evidence: baseline_Screening_Log.csv;Method_and_Deviations;rameen/novelty_scope_notes.md

Next: Use Remaining_559_Screening_Queue.csv for two actual votes, abstract retrieval where needed, full-text attempts and IC/EC decisions. Otherwise submit a clearly bounded first draft and retain the deviation; instructor acceptance has not been evidenced.

MA05 Abstract coverage (Explicit limitation)

Frozen canonical register: 2933 without abstracts and 1142 with abstracts. Current 559-row queue separately audits recoverable metadata; availability does not prove reading.

Evidence: baseline_Screening_Log.csv;Abstract available; integrated source-review registers / remaining_559_audit.json; integrated source-review registers / remaining_559_audit.json

Next: Use actual source abstracts where recoverable; request abstract/full text for uncertain records; exclude only on a supported IC/EC reason. Title-only decisions remain a limitation. See remaining_559_audit.json.

MA06 Independent review (Explicit limitation)

All 200 included-study QA items and 52 outcome rows have source-review entries; separate named human votes are absent.

Evidence: Submittedprotocolsection9;teamworkdivision;reviewregisters; integrated source-review registers / remaining_559_audit.json; integrated source-review registers / remaining_559_audit.json

Next: Use teammate rotation; preserve initial independent judgments, real names/times, disagreements and adjudication. Do not copy prepared appraisals into independent-vote fields.

MA07 Shared search provenance (Explicit limitation)

IEEE/Springer shared searches are attributed. Historical exact execution times and complete operator receipts remain unavailable.

Evidence: baseline_Search_Log.csv;IMPORT_PROVENANCE.md

Next: Disclose reuse and missing times; recorded dates do not establish within-day times. No retrospective rerun is claimed.

MA08 ACM provenance (Explicit limitation)

ACM numbered files contribute 1582 active occurrences. Two master BibTeX files parse to 1829 entries and are not active additional occurrences. Display1600/claimed1603 versus parsed1829 and exact run mapping remain unresolved.

Evidence: baseline_Search_Log.csv;ACM-MASTER;ACM-NUMBERED-UNMAPPED

Next: Retain count fields separately; do not add master records to PRISMA. Verify raw-file mapping with original collector if available.

MA09 ScienceDirect provenance (Explicit limitation)

ScienceDirect active register has1898 occurrences. MPS displayed517 versus registered563 remains unresolved. A historical full-export manifest exists, but its named archive files are absent from this checkout; four standalone samples do not replace the full archive.

Evidence: baseline_Search_Log.csv;Method_and_Deviations

Next: Preserve manifest as historical evidence; do not assert current byte verification or export completeness. Missing archive and field/facet uncertainty remain disclosed.

MA10 Search fields and synonyms (Explicit limitation)

Accepted query/field deviations from the submitted protocol are documented. Broad indexed fields and missing run/facet mapping cannot be reconstructed from reading copies.

Evidence: baseline_Search_Log.csv;submittedprotocolsearchstrings

Next: Report actual preserved strings; distinguish planned versus executed searches and supplementary retrieval. No unseen run or missing filter is invented.

MA11 Source versions (Explicit limitation)

All20 study copies plus REEF companion are inspected. iGniter final copy is checked; REEF E018-E026 belong to conferenceP005; Tally is pinned to arXivv3. ParvaGPU final-text equivalence remains unverified.

Evidence: eshaal/sources;raazia/README.md; integrated source-review registers / remaining_559_audit.json; integrated source-review registers / remaining_559_audit.json

Next: Retain ParvaGPU manuscript-only quantitative status and IC1 version condition; no false final-version certification.

MA12 QA consistency (Resolution documented)

Claim-specific Q9 calibration and applicable-denominator review are applied across20 forms. P003 artifact appendix supports Q10=2. Original scores remain frozen and item-level changes are logged.

Evidence: QA_Source_Review.csv;submittedprotocolsection10; integrated source-review registers / remaining_559_audit.json; integrated source-review registers / remaining_559_audit.json

Next: Use integrated current scores; human calibration/signoff remains a separate pending step. N/A only P001Q7 and P006Q9.

MA13 Extraction (Resolution documented)

All52 retained outcomes source-checked across the three packages; corrected anchors, metric meanings and source-version boundaries integrated.

Evidence: Outcome_Source_Review.csv; integrated source-review registers / remaining_559_audit.json; integrated source-review registers / remaining_559_audit.json

Next: Do not treat reporting verification as experiment replication; named independent checks remain pending.

MA14 Minimum-study instruction (Resolution documented)

User clarified Sir requires20 studies; docs/guidelines.md states minimum15to20. Corpus has20 unique studies/21reports because REEF conference+journal is one study.

Evidence: TEAM_WORK_DIVISION.md;POST_TASK2_CHECKLIST.md;userinstruction

Next: No count-driven extra studies are required for this draft. If completing the retained queue identifies more eligible studies, include them under the predefined criteria rather than stopping at20.

Independent-check return

Rameen independently checks P001–P007 plusP023. Eshaal independently checks Raazia'sP008–P014. Enter the actual reviewer,checked time,initial decision,source anchor and disagreements separately. These prepared review records cannot be copied into human-vote columns as independent signoff.

Checked: 2026-10-22T14:54:10.997021+05:00. Attribution: AI assistant preparing Eshaal assignment at Raazia request. Full texts are excluded from public Git; source hashes and appraisal facts are retained.